

Fabrikam Service Insights & Automation Platform Solution Design Document.

1. Problem Understanding

Business Context

Fabrikam Consulting supports a mid-sized enterprise using Dynamics Customer Service. Leadership currently lacks real-time visibility, automation mechanisms, and structured analytical reporting to support operational and strategic decision-making.

Current Challenges Identified

Problem	Business Impact
Static CRM Views	No dynamic reporting or real-time visibility
No trend Insight	Leadership cannot forecast workload or SLA risks
Late SLA breach detection	Customer dissatisfaction and compliance risk
Attachments stored in CRM	Increased CRM storage costs
No automation monitoring	System failures go unnoticed

2. Assumptions

- Case data includes SLA due dates and resolution timestamps.
- Power BI will be used for reporting and dashboards.
- Power Automate is available for workflow automation.
- Azure Active Directory manages authentication.
- Dynamics Customer Service data can be extracted to Azure SQL.

3. High-Level Solution Architecture (Conceptual)

Proposed Architecture Overview

The solution will consist of:

1. Dynamics CRM as source system
2. Azure SQL as reporting database
3. Power BI for reporting and analytics
4. Power Automate for SLA breach notification
5. Azure AD for authentication and role-based access

4. Data Model Design for Operational Reporting

4.1 Identify Required Data Fields

- Active cases monitoring
- SLA Performance
- Trend analysis

TABLE: Cases

Column Name	Data Type	Purpose
CaseID	INT	Unique case identifier
CreatedDate	DATETIME	When case was created
ClosedDate	DATETIME	When case was resolved
Priority	VARCHAR	Low/Medium/High
Status	VARCHAR	Open/In progress/Closed
AssignedTo	VARCHAR	Case owner
SLA_DueDate	DATETIME	SLA deadline
CustomerID	INT	Linked customers

Derived Columns (Reporting Logic)

- $\text{ResolutionTime} = \text{ClosedDate} - \text{CreatedDate}$

- SLA_Breached = IF(ClosedDate > SLA_DueDate, Yes, No)
- ActiveCaseFlag = IF(Status != "Closed", 1, 0)

4.2 *Optional Supporting Table*

TABLE: SLA_Config

Column	Purpose
Priority	Priority Level
SLA_Hours	Allowed SLA duration

This allows flexibility instead of hardcoding SLA.

Justification:

Separating SLA rules into a configuration table ensures maintainability and avoids hard-coded business logic.

5. KPI Definitions & Business Logic

1. Total Cases

COUNT(CaseID)

2. Active Cases

COUNT(CaseID where status! = Closed)

3. SLA Breach Percentage

(Number of Breached Cases/Total Cases) *100

4. Average Resolution Time

AVG(ResolutionTime)

5. Case Volume Trend

COUNT(CaseID) grouped by month